

Introduction to Abstract Algebra III: Homework 5

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Exercise 43.30. Show that the only $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ are $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{6})$, and $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. [Hint: Show that a subfield of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ that is a degree 2 extension of $\mathbb{Q}$ must be of the form $\mathbb{Q}(\sqrt{s})$ for some rational number s .]

Solution. We first show that any subfield of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ that is a degree 2 extension of $\mathbb{Q}$ must be of the form $\mathbb{Q}(\sqrt{s})$ for some rational number s . Let E be a subfield of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ such that $[E : \mathbb{Q}] = 2$. Since E is a degree 2 extension of $\mathbb{Q}$, there exists an element $\alpha \in E \setminus \mathbb{Q}$ such that $E = \mathbb{Q}(\alpha)$. Since $\alpha \in \mathbb{Q}(\sqrt{2}, \sqrt{3})$, we can write:

$$\alpha = a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6},$$

for some $a, b, c, d \in \mathbb{Q}$. Replacing α by $\alpha - a$, we may assume $a = 0$ without loss of generality, so $\alpha = b\sqrt{2} + c\sqrt{3} + d\sqrt{6}$. We compute:

$$\alpha^2 = 2b^2 + 3c^2 + 6d^2 + 2bc\sqrt{6} + 2bd\sqrt{12} + 2cd\sqrt{18}.$$

Simplifying, $\sqrt{12} = 2\sqrt{3}$ and $\sqrt{18} = 3\sqrt{2}$, so:

$$\alpha^2 = (2b^2 + 3c^2 + 6d^2) + (6cd)\sqrt{2} + (4bd)\sqrt{3} + (2bc)\sqrt{6}.$$

For α to generate a degree 2 extension of $\mathbb{Q}$, its minimal polynomial over $\mathbb{Q}$ must be $x^2 - s$ for some $s \in \mathbb{Q}$, which requires $\alpha^2 \in \mathbb{Q}$. Thus all irrational parts must vanish:

$$6cd = 0, \quad 4bd = 0, \quad 2bc = 0.$$

These conditions imply that at most one of b, c, d is nonzero. Since $\alpha \notin \mathbb{Q}$, at least one of b, c, d is nonzero, so exactly one is nonzero. In each case we get $\alpha = b\sqrt{2}$, $\alpha = c\sqrt{3}$, or $\alpha = d\sqrt{6}$, giving $E = \mathbb{Q}(\sqrt{2})$, $E = \mathbb{Q}(\sqrt{3})$, or $E = \mathbb{Q}(\sqrt{6})$, respectively. In each case $E = \mathbb{Q}(\sqrt{s})$ for $s = 2, 3, 6 \in \mathbb{Q}$.

Now, we show that the only subfields of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ are $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{6})$, and $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. Since $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ is the splitting field of $(x^2 - 2)(x^2 - 3)$ over $\mathbb{Q}$, it is a Galois extension of $\mathbb{Q}$ of degree 4. The Galois group $G = G(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q})$ has order 4 and consists of the automorphisms determined by:

$$\begin{aligned} \sigma_1 : \sqrt{2} &\mapsto \sqrt{2}, & \sqrt{3} &\mapsto \sqrt{3} \\ \sigma_2 : \sqrt{2} &\mapsto -\sqrt{2}, & \sqrt{3} &\mapsto \sqrt{3} \\ \sigma_3 : \sqrt{2} &\mapsto \sqrt{2}, & \sqrt{3} &\mapsto -\sqrt{3} \\ \sigma_4 : \sqrt{2} &\mapsto -\sqrt{2}, & \sqrt{3} &\mapsto -\sqrt{3}. \end{aligned}$$

Since every non-identity element has order 2, we have $G \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, which has exactly three subgroups of order 2:

$$H_1 = \{\sigma_1, \sigma_2\}, \quad H_2 = \{\sigma_1, \sigma_3\}, \quad H_3 = \{\sigma_1, \sigma_4\},$$

in addition to the trivial subgroup $\{\sigma_1\}$ and the whole group G . By the Galois correspondence, the five subgroups correspond to five subfields: the whole group G corresponds to $\mathbb{Q}$, the three subgroups of order 2 correspond to the fixed fields $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{2})$, and $\mathbb{Q}(\sqrt{6})$, and the trivial subgroup corresponds to $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. By the first part of this solution, there are no degree 2 subfields other than these three, and clearly there are no subfields of degree 3 since it doesn't divide 4. Thus, the only subfields of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ are $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{6})$, and $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. $\square$

Exercise 43.32. Let $F(\alpha_1, \dots, \alpha_n)$ be an extension field of F . Show that any automorphism σ of $F(\alpha_1, \dots, \alpha_n)$ leaving F fixed is completely determined by the n values $\sigma(\alpha_i)$.

Solution. Let σ be an automorphism of $F(\alpha_1, \dots, \alpha_n)$ leaving F fixed. Since σ is a field homomorphism fixing F , we have $\sigma(a) = a$ for all $a \in F$. We claim that for any rational expression $f(\alpha_1, \dots, \alpha_n)$ with coefficients in F , we have:

$$\sigma(f(\alpha_1, \dots, \alpha_n)) = f(\sigma(\alpha_1), \dots, \sigma(\alpha_n)).$$

Indeed, since σ preserves addition and multiplication, a straightforward induction on the structure of polynomial expressions gives $\sigma(p(\alpha_1, \dots, \alpha_n)) = p(\sigma(\alpha_1), \dots, \sigma(\alpha_n))$ for any polynomial p with coefficients in F . Since σ is a field automorphism it also preserves division, so for any $f = p/q$ with $q(\alpha_1, \dots, \alpha_n) \neq 0$ we have:

$$\sigma\left(\frac{p(\alpha_1, \dots, \alpha_n)}{q(\alpha_1, \dots, \alpha_n)}\right) = \frac{\sigma(p(\alpha_1, \dots, \alpha_n))}{\sigma(q(\alpha_1, \dots, \alpha_n))} = \frac{p(\sigma(\alpha_1), \dots, \sigma(\alpha_n))}{q(\sigma(\alpha_1), \dots, \sigma(\alpha_n))}.$$

Thus, the value of σ on any element of $F(\alpha_1, \dots, \alpha_n)$ is completely determined by the n values $\sigma(\alpha_i)$. $\square$

Exercise 43.33. Let E be an algebraic extension of a field F , and let σ be an automorphism of E leaving F fixed. Let $\sigma \in E$. Show that σ induces a permutation of the set of all zeros of $\text{irr}(\alpha, F)$ that are in E .

Solution. Let $f(x) = \text{irr}(\alpha, F)$ and let $S = \{\beta \in E : f(\beta) = 0\}$ be the set of zeros of $f(x)$ in E . Since f has finite degree, S is a finite set. We first show that σ maps S into S . If $\beta \in S$, then since the coefficients of $f(x)$ lie in F and σ fixes every element of F , we have:

$$f(\sigma(\beta)) = \sigma(f(\beta)) = \sigma(0) = 0,$$

so $\sigma(\beta) \in S$. Thus σ restricts to a map $\sigma|_S : S \rightarrow S$. Since σ is a field automorphism, it is injective, so $\sigma|_S$ is an injective self-map of the finite set S . Any injective self-map of a finite set is a bijection, so $\sigma|_S$ is a permutation of S . $\square$

Exercise 43.34. Let E be an algebraic extension of a field F . Let $S = \{\sigma_i | i \in I\}$ be a collection of automorphisms of E such that every σ_i leaves each element of F fixed. Show that if S generates the subgroup H of $G(E/F)$, then $E_S = E_H$.

Solution. By definition, E_S is the fixed field of the collection S , which consists of all elements of E that are fixed by every $\sigma_i \in S$. On the other hand, E_H is the fixed field of the subgroup H generated by S , which consists of all elements of E that are fixed by every element of H . Since H is generated by S , every element of H can be expressed as a finite composition of elements from S . Therefore, if an element of E is fixed by every $\sigma_i \in S$, it must also be fixed by every element of H , since each element of H is a composition of elements from S . Conversely, if an element of E is fixed by every element of H , it must be fixed by every $\sigma_i \in S$, since each σ_i is an element of H . Thus, the sets of elements fixed by S and by H are the same, and we conclude that $E_S = E_H$. $\square$

Exercise 43.36. Referring to Exercise 35, let F be a finite field of characteristic p , and let φ be the Frobenius automorphism on F . Prove that the fixed field $F_{\{\varphi\}}$ is isomorphic with $\mathbb{Z}_p$.

Solution. We first show that $\mathbb{Z}_p \subseteq F_{\{\varphi\}}$. For any $a \in \mathbb{Z}_p$, we have:

$$\varphi(a) = a^p.$$

Since a is an integer between 0 and $p-1$, we can use Fermat's little theorem, which states that if p is a prime and a is an integer not divisible by p , then $a^{p-1} \equiv 1 \pmod{p}$. If $a = 0$, then $\varphi(0) = 0^p = 0$. If $a \neq 0$, then:

$$a^p = a^{(p-1)+1} = a^{p-1} \cdot a \equiv 1 \cdot a = a \pmod{p}.$$

Thus, in either case, we have $\varphi(a) = a$, so every element of $\mathbb{Z}_p$ is fixed by φ . Therefore, $\mathbb{Z}_p \subseteq F_{\{\varphi\}}$.

Now, we show that $F_{\{\varphi\}} \subseteq \mathbb{Z}_p$. Suppose $\alpha \in F_{\{\varphi\}}$, so $\varphi(\alpha) = \alpha^p = \alpha$. This means that α is a root of the polynomial $x^p - x \in \mathbb{Z}_p[x]$. Since $x^p - x$ has degree p , it has at most p roots in F . But from the first part, every element of $\mathbb{Z}_p$ satisfies $a^p = a$, so the p elements of $\mathbb{Z}_p$ are already p distinct roots of $x^p - x$. Therefore, the set of roots of $x^p - x$ in F is exactly $\mathbb{Z}_p$, and so $\alpha \in \mathbb{Z}_p$. Thus $F_{\{\varphi\}} \subseteq \mathbb{Z}_p$.

Combining both directions, we have $F_{\{\varphi\}} = \mathbb{Z}_p$, and so $F_{\{\varphi\}}$ is isomorphic with $\mathbb{Z}_p$. $\square$

Exercise 43.37. Referring to Exercise 35, show that the finite assumption is necessary by finding an example of a field F with characteristic p , such that the map $\varphi : F \rightarrow F$ given by $\varphi(\alpha) = \alpha^p$ is not an automorphism.

Solution. Take $F = \mathbb{Z}_p(t)$, the field of rational functions in one variable t over $\mathbb{Z}_p$. The characteristic of F is p , since it contains $\mathbb{Z}_p$ as a subfield. The map $\varphi : F \rightarrow F$ given by $\varphi(\alpha) = \alpha^p$ is not an automorphism because it is not surjective. We claim that $t \in \mathbb{Z}_p(t)$ has no preimage under φ . Suppose for contradiction that $\varphi(f/g) = t$ for some $f, g \in \mathbb{Z}_p[t]$, so:

$$\left(\frac{f}{g}\right)^p = t \Rightarrow f^p = t \cdot g^p.$$

The left side has degree $p \cdot \deg(f)$, which is divisible by p . However, the right side has degree $p \cdot \deg(g) + 1$, which is not divisible by p . This is a contradiction. Therefore, t is not in the image of φ , so φ is not surjective and hence not an automorphism of F . $\square$

Exercise 43.38. We saw in Corollary 28.18 that the cyclotomic polynomial

$$\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \cdots + x + 1,$$

is irreducible over $\mathbb{Q}$ for every prime p . Let ζ be a zero of $p(x)$, and consider the field $\mathbb{Q}(\zeta)$.

- (i) Show that $\zeta, \zeta^2, \dots, \zeta^{p-1}$ are distinct zeros of $p(x)$, and conclude that they are all the zeros of $p(x)$.
- (ii) Deduce from Corollary 43.19 and part (i) of this exercise that $G(\mathbb{Q}(\zeta)/\mathbb{Q})$ is abelian of order $p - 1$.
- (iii) Show that the fixed field of $G(\mathbb{Q}(\zeta)/\mathbb{Q})$ is $\mathbb{Q}$. [Hint: Show that

$$\{\zeta, \zeta^2, \dots, \zeta^{p-1}\},$$

is a basis for $\mathbb{Q}(\zeta)$ over $\mathbb{Q}$, and consider which linear combinations of $\zeta, \zeta^2, \dots, \zeta^{p-1}$ are fixed by all elements of $G(\mathbb{Q}(\zeta)/\mathbb{Q})$].

Solution to (i). Since ζ is a zero of $\Phi_p(x)$, we have $\zeta^p = 1$ and $\zeta \neq 1$. Thus, ζ is a primitive p -th root of unity. The distinct powers $\zeta, \zeta^2, \dots, \zeta^{p-1}$ are all roots of $\Phi_p(x)$ because:

$$\Phi_p(\zeta^k) = \frac{(\zeta^k)^p - 1}{\zeta^k - 1} = \frac{\zeta^{kp} - 1}{\zeta^k - 1} = \frac{1 - 1}{\zeta^k - 1} = 0,$$

for $k = 1, 2, \dots, p-1$. Since $\Phi_p(x)$ has degree $p-1$, it can have at most $p-1$ distinct roots. Therefore, the roots $\zeta, \zeta^2, \dots, \zeta^{p-1}$ are all the zeros of $\Phi_p(x)$. $\square$

Solution to (ii). Let $E = G(\mathbb{Q}(\zeta)/\mathbb{Q})$. Then, since $\Phi_p(x)$ is irreducible over $\mathbb{Q}$, the extension $\mathbb{Q}(\zeta)/\mathbb{Q}$ is a Galois extension of degree $p-1$. Then, for any $\sigma_j, \sigma_m \in E$, we have $\sigma_j(\zeta) = \zeta^j$ and $\sigma_m(\zeta) = \zeta^m$ for some integers j, m coprime to p . The composition $\sigma_j \circ \sigma_m$ acts on ζ as follows:

$$(\sigma_j \circ \sigma_m)(\zeta) = \sigma_j(\sigma_m(\zeta)) = \sigma_j(\zeta^m) = (\zeta^m)^j = \zeta^{mj}.$$

Similarly, the composition $\sigma_m \circ \sigma_j$ acts on ζ as:

$$(\sigma_m \circ \sigma_j)(\zeta) = \sigma_m(\sigma_j(\zeta)) = \sigma_m(\zeta^j) = (\zeta^j)^m = \zeta^{jm}.$$

Since $mj = jm$, we have $(\sigma_j \circ \sigma_m)(\zeta) = (\sigma_m \circ \sigma_j)(\zeta)$, and since ζ generates $\mathbb{Q}(\zeta)$, it follows that $\sigma_j \circ \sigma_m = \sigma_m \circ \sigma_j$. Therefore, E is abelian. Moreover, since E is the Galois group of a degree $p-1$ extension, we have $|E| = p-1$. $\square$

Solution to (iii). Since $\mathbb{Q}(\zeta)/\mathbb{Q}$ is a Galois extension of degree $p - 1$ with Galois group $G = G(\mathbb{Q}(\zeta)/\mathbb{Q})$ of order $p - 1$, the fundamental theorem of Galois theory tells us that the fixed field of G satisfies:

$$[\mathbb{Q}(\zeta) : F_G] = |G| = p - 1.$$

Since $\mathbb{Q} \subseteq F_G \subseteq \mathbb{Q}(\zeta)$ and $[\mathbb{Q}(\zeta) : \mathbb{Q}] = p - 1$, we have:

$$[\mathbb{Q}(\zeta) : \mathbb{Q}] = [\mathbb{Q}(\zeta) : F_G][F_G : \mathbb{Q}] \Rightarrow p - 1 = (p - 1)[F_G : \mathbb{Q}],$$

which gives $[F_G : \mathbb{Q}] = 1$, and therefore $F_G = \mathbb{Q}$. Thus, the fixed field of $G(\mathbb{Q}(\zeta)/\mathbb{Q})$ is $\mathbb{Q}$. $\square$

Exercise 44.6. Find the degree over $\mathbb{Q}$ of the splitting field over $\mathbb{Q}$ of the polynomial $(x^2 - 2)(x^3 - 2)$ in $\mathbb{Q}[x]$.

Solution. The splitting field of $(x^2 - 2)(x^3 - 2)$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt{2}, \sqrt[3]{2}, \omega)$, where $\omega = \frac{-1+i\sqrt{3}}{2}$ is a primitive cube root of unity. This is because $x^2 - 2$ has roots $\pm\sqrt{2}$ and $x^3 - 2$ has roots $\sqrt[3]{2}, \omega\sqrt[3]{2}, \omega^2\sqrt[3]{2}$, all of which lie in $\mathbb{Q}(\sqrt{2}, \sqrt[3]{2}, \omega)$. We compute the degree using the tower:

$$\mathbb{Q} \subseteq \mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{Q}(\sqrt[3]{2}, \omega) \subseteq \mathbb{Q}(\sqrt[3]{2}, \omega, \sqrt{2}).$$

By Eisenstein's criterion, taking $p = 3$, we know that $x^3 - 2$ is irreducible over $\mathbb{Q}$. So, $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$. Furthermore, ω satisfies $x^2 + x + 1 = 0$, which is irreducible over $\mathbb{Q}$ and has no roots in $\mathbb{Q}(\sqrt[3]{2})$. Thus, $[\mathbb{Q}(\sqrt[3]{2}, \omega) : \mathbb{Q}(\sqrt[3]{2})] = 2$. The value $\sqrt{2}$ is a root of $x^2 - 2$. We know that $[\mathbb{Q}(\sqrt[3]{2}, \omega) : \mathbb{Q}] = 6$ and $[\mathbb{Q}(\sqrt[3]{2}, \omega, \sqrt{2}) : \mathbb{Q}]$ must be divisible by both 6 and 2 = $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}]$. However, $\sqrt{2} \notin \mathbb{Q}(\sqrt[3]{2}, \omega)$. By the tower law, we have:

$$[\mathbb{Q}(\sqrt{2}, \sqrt[3]{2}, \omega) : \mathbb{Q}] = 3 \cdot 2 \cdot 2 = 12. \quad \square$$

Exercise 44.8. What is the order of $G(\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})/\mathbb{Q})$?

Solution. The order of $G(\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})/\mathbb{Q})$ equals the degree $[\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3}) : \mathbb{Q}]$, since $\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})$ is the splitting field of the separable polynomial $x^3 - 2$ over $\mathbb{Q}$ and is therefore a Galois extension. We note that $\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3}) = \mathbb{Q}(\sqrt[3]{2}, \omega)$, where $\omega = \frac{-1+i\sqrt{3}}{2}$ is a primitive cube root of unity, since $i\sqrt{3} = 2\omega + 1$. We compute the degree using the tower:

$$\mathbb{Q} \subseteq \mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{Q}(\sqrt[3]{2}, \omega).$$

First, $x^3 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein's criterion, so $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$. Second, ω satisfies $x^2 + x + 1 = 0$, which is irreducible over $\mathbb{Q}$ and has no roots in $\mathbb{Q}(\sqrt[3]{2})$ since $\mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{R}$, so $[\mathbb{Q}(\sqrt[3]{2}, \omega) : \mathbb{Q}(\sqrt[3]{2})] = 2$. By the tower law:

$$|G(\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})/\mathbb{Q})| = [\mathbb{Q}(\sqrt[3]{2}, \omega) : \mathbb{Q}] = 3 \cdot 2 = 6. \quad \square$$

Exercise 44.10. Let α be a zero of $x^3 + x^2 + 1$ over $\mathbb{Z}_2$. Show that $x^3 + x^2 + 1$ splits in $\mathbb{Z}_2(\alpha)$. [Hint: There are eight elements in $\mathbb{Z}_2(\alpha)$. Exhibit two more zeros of $x^3 + x^2 + 1$, in addition to α , among these eight elements. Alternatively, use the results of Section 42.]

Solution. We will use Galois correspondence to show that $x^3 + x^2 + 1$ splits over $\mathbb{Z}_2(\alpha)$. We do this by first showing that $\mathbb{Z}_2(\alpha) \cong GF(p^n)$, and then showing that $GF(p^n)$ contains all the roots of $x^3 + x^2 + 1$. We have $p = 2$ and $n = 3$, so $GF(2^3) = GF(8)$. Since $f(x) = x^3 + x^2 + 1$ is irreducible over $\mathbb{Z}_2$ ($f(x)$ has no roots in $\mathbb{Z}_2$), the extension $\mathbb{Z}_2(\alpha)$ has degree 3 over $\mathbb{Z}_2$. Then, there exists a unique field homomorphism $\mathbb{Z}_2(\alpha)$ to $GF(8)$ that sends α to a root of $f(x)$ in $GF(8)$. Since $\mathbb{Z}_2(\alpha)$ and $GF(8)$ are both fields of order 8, this homomorphism is an isomorphism. Thus, $\mathbb{Z}_2(\alpha) \cong GF(8)$. We also proved in HW 3 that $x^{p^n} - x$ is the product of all monic irreducible polynomials over $\mathbb{Z}_p$ whose degree divides n . In particular, since $f(x)$ is an irreducible polynomial of degree 3, it must divide $x^8 - x$ in $\mathbb{Z}_2[x]$. Therefore, all the roots of $f(x)$ are also roots of $x^8 - x$. Since $x^8 - x$ splits completely over $GF(8)$, it follows that $f(x)$ also splits completely over $GF(8)$. Since $\mathbb{Z}_2(\alpha) \cong GF(8)$, we conclude that $f(x)$ splits in $\mathbb{Z}_2(\alpha)$. $\square$